A Case Study Report

on

“Case Studies in Data Science: Clustering of Online Retail Baskets and Prediction of Student Dropout”

(Case Study 1 and Case Study 4)

for the course

Fundamentals of Data Science (22CSC59)

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING

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COMPUTER SCIENCE AND ENGINEERING

by

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Abstract

This report combines two case studies prepared for the course Fundamentals of Data Science. Both follow the same workflow of data audit, preparation, exploration, modelling and honest evaluation, but they address different problems: one finds structure in unlabelled retail transactions, and the other predicts the outcome of university students.

Case Study 1: Shopping-Basket Missions and Atypical Baskets

This study converts positional transaction records into clean product sets and binary basket vectors, summarizes basket size and product support, reduces the feature space with PCA, and compares K-means, Ward hierarchical clustering, DBSCAN, and spectral clustering. K-means supplies a complete five-group basket taxonomy, whereas DBSCAN isolates 17.56% of transactions as noise. The strongest core-only validation scores therefore come with a coverage trade-off. Product profiles reveal unusually sharp anchor-product effects, so the clusters are treated as hypotheses for validation rather than proven customer segments.

Case Study 4: Predicting Student Dropout and Academic Success

This study analyses 4,424 undergraduate students labelled Dropout, Enrolled or Graduate. After auditing the data and categorising attributes by the meaning of their encoded values, exploratory analysis relates the outcome to admission grade, first-semester performance, age at enrolment and financial indicators. A Kruskal-Wallis test, chosen because the assumptions of ANOVA fail, shows that first-semester grades differ significantly across outcomes with a large effect size. A multinomial logistic regression built from early information reaches 75.0% accuracy and a macro F1-score of 0.669, performing well for Graduate and Dropout students but poorly for Enrolled students. Limitations, recommendations and conditions for responsible use are discussed, and associations are distinguished from causation throughout.

A comparative summary of both studies, their common lessons and a combined reference list are given at the end of the report.

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CASE STUDY 1

"Shopping-Basket Missions and Atypical Baskets: A Comparative Clustering Analysis of Online Retail Transactions"

Fundamentals of Data Science (22CSC59)

Sriya Ammanamanchi (160124733156) | Section: CSE-3 | Year: 2026-27

Case Study at a Glance

Dataset	7,500 online-retail transactions (20 positional product columns, 119 distinct products)
Techniques	Pandas cleaning, binary basket vectors, PCA (20 components), K-means, Ward hierarchical clustering, DBSCAN and spectral clustering
Evaluation	Silhouette coefficient, Davies–Bouldin index and coverage
Headline result	K-means (K = 5) gives a complete five-group basket taxonomy; DBSCAN scores best on core points but labels 17.56% of transactions as noise

1. PROBLEM STATEMENT

1.1 Problem Definition, Research Questions and Objectives

The online retail store possesses transaction records containing multiple products purchased together, but it has no systematic description of transaction characteristics or groups of similar shopping baskets. The business problem is to transform these unstructured basket records into reproducible statistical evidence and to assess whether unsupervised learning can produce useful transaction groups.

The study addresses nine linked tasks: understand the data structure; clean records with Pandas; calculate descriptive statistics and frequent products; visualize support and basket-size distributions; create numerical basket vectors; apply PCA; implement four clustering algorithms; compare silhouette and Davies–Bouldin results; and derive limitations and business implications.

Distinctive angle

The central question is not simply "which algorithm has the highest score?" A high score can conceal excluded observations, tiny clusters, or representation choices. Accordingly, the report compares two operational uses:

- **Complete basket taxonomy:** can every transaction be assigned to a compact and interpretable set of recurring compositions?
- **Atypical-basket detection:** can a density-based method separate common structure from unusual transactions, and how much coverage is sacrificed?

This makes coverage, balance, and product profiles part of model assessment alongside the required indices. The analytical theme is to distinguish useful basket-composition patterns from algorithmic artefacts by combining internal validation, coverage, and product-level interpretation.

1.2 Scope and Unit of Analysis

Each row is one transaction. The data do not contain customer identifiers, timestamps, quantities, prices, costs, or margins. The report therefore does not claim to segment customers, forecast demand, estimate revenue, or establish causal cross-selling effects. "Frequency" means the number of baskets containing a product, not units sold.

2. METHODOLOGY

2.1 Data Audit and Cleaning

The CSV contains 7,500 rows and 20 columns. Inspection shows that the header labels are products rather than meaningful field names: the columns are positional item slots. Later columns have increasing numbers of blanks because shorter baskets leave positions unused. Consequently, conventional column-wise missing-value imputation would fabricate products and is inappropriate.

For every row, null positions are removed, product strings are trimmed, internal whitespace is standardized, and case is normalized. Duplicate mentions are removed only within a basket. Repeated baskets across rows remain because repeated transactions are genuine frequency evidence. Assertions confirm 7,500 non-empty cleaned baskets. The companion workbook was also parsed as comma-separated basket rows and matched the CSV transaction-for-transaction.

Table 1.1: Data-structure decisions and their consequences

Observed issue	Decision	Reason
Twenty product-labelled columns	Treat columns as positions	Headers are not transaction attributes
Unequal null counts	Drop null positions within rows	Blanks denote shorter baskets
Product strings	Trim, lowercase, normalize spaces	Prevent spelling variants from splitting support
Repeated basket patterns	Retain across transactions	Recurrence carries useful frequency information
No price, time, customer ID	Restrict claims to transactions	Avoid unsupported financial or customer inference

2.2 Descriptive Analysis and Visualization

Basket size is the number of distinct cleaned products in a transaction. Product support is its basket occurrence count. The report examines distributional summaries, the top products, singleton and large-basket shares, and the concentration of the ten most frequent items. Histograms and horizontal support bars are used because they preserve the long tail and make product labels readable.

2.3 Binary Representation and PCA

A multi-label binarizer converts the baskets to a 7,500 × 119 matrix X, where $x_{ij} = 1$ if product j occurs in basket i. The matrix density is 0.0329, confirming substantial sparsity. Binary presence is appropriate for co-occurrence but discards quantity and value.

PCA produces 20 components without variance scaling. The unstandardized representation preserves high-variance product/co-occurrence directions. These 20 scores are used for K-means, Ward, and spectral clustering. DBSCAN instead uses z-standardized PC scores so that its fixed neighbourhood radius is not dominated only by the highest-variance component. The first two PCs are used for visualization only.

2.4 Clustering Design and Parameter Choices

K-means (MacQueen, 1967) is scanned for K = 2,…,8 with 20 initializations. K = 5 is selected as an operational compromise rather than a claim of five mathematically proven natural groups. Ward hierarchical clustering uses the same K and full data. DBSCAN (Ester et al., 1996) uses

eps = 3.5 and min_samples = 15 after neighbourhood diagnostics; it determines cluster count itself and can label noise. Spectral clustering (Ng et al., 2002) uses a 20-nearest-neighbour affinity graph and five clusters on a fixed 2,000-transaction evaluation sample, avoiding a costly full affinity matrix.

2.5 Evaluation Protocol

A fixed random sample of 2,000 transactions supports comparable calculation. Silhouette coefficient (Rousseeuw, 1987) measures cohesion relative to separation and is higher-is-better. Davies–Bouldin index (DBI; Davies and Bouldin, 1979) compares within-cluster scatter with between-cluster separation and is lower-is-better. DBSCAN is scored only on non-noise sample points, and coverage is reported separately. This distinction is essential: excluding difficult points can improve geometry without providing a complete segmentation.

Fair-comparison caveat. The experiment is diagnostic, not a perfectly controlled tournament. Spectral clustering is fitted on the evaluation sample, DBSCAN uses standardized PCs and rejects noise, and the other methods use unstandardized PCs with full coverage. Conclusions therefore combine scores with operational suitability.

3. STATISTICAL FINDINGS

3.1 Frequently Occurring Products

Mineral water occurs in 1,787 baskets, followed by eggs (1,348), spaghetti (1,306), french fries (1,282), and chocolate (1,229). These products are candidates for high-impact basket anchors. However, the analysis does not observe price or margin, so frequency cannot be equated with commercial importance.

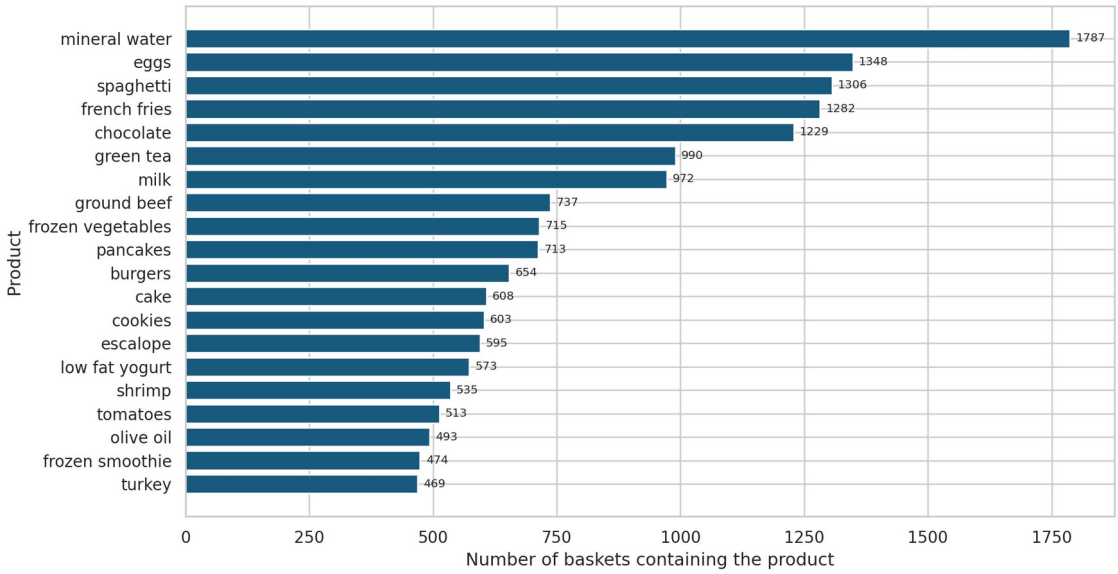


Figure 1.1: Top 20 products by number of baskets containing each item

Interpretation. Support declines from a dominant core to a long tail. Common products can dominate Euclidean geometry and cluster formation; rare products, in contrast, may have high lift but too little support for reliable recommendations. Both prevalence and lift are therefore used in cluster profiling.

3.2 Transaction-Size Characteristics

The cleaned data contain 29,338 item occurrences across 7,500 baskets and 119 products. Mean basket size is 3.912, but the median is 3 and the maximum is 19. The mean–median gap and standard deviation of 2.899 indicate right skew.

Table 1.2: Basket-level descriptive findings

Measure	Result	Interpretation
Mean / median basket size	3.912 / 3	Large-basket tail raises the mean
25th / 75th percentile	2 / 5	Half of baskets contain 2–5 products
Single-item baskets	23.39%	A large low-complexity transaction component
Baskets with at most two items	41.49%	Small missions are common
Baskets with at least ten items	5.25%	Large shops are a minority tail
Top-ten occurrence share	37.76%	High-support products exert strong influence

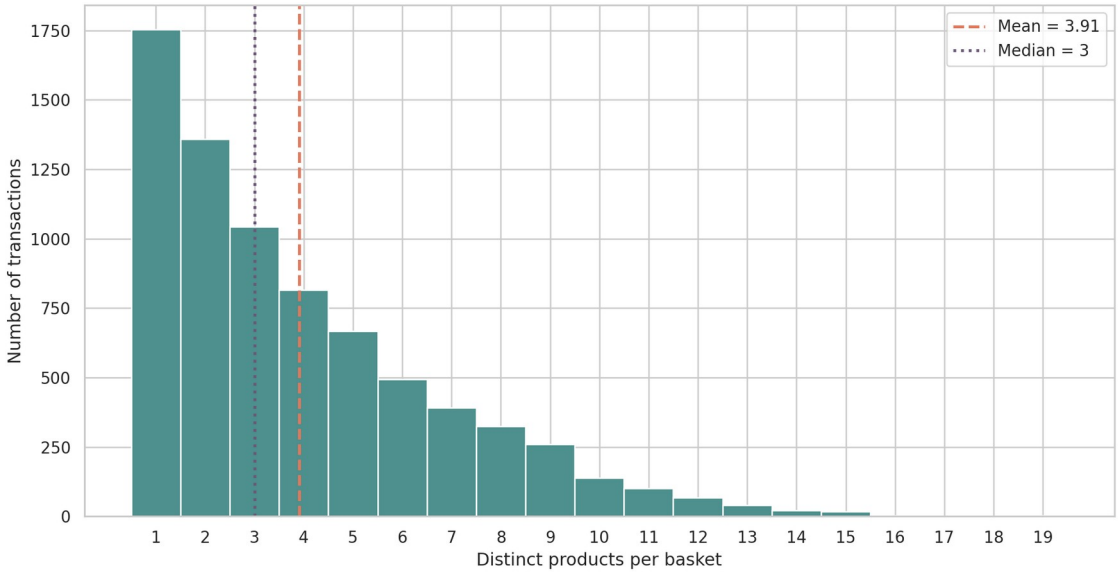


Figure 1.2: Distribution of distinct products per transaction

Interpretation. Most transactions are compact, while a small tail reaches 19 items. Operationally, a single "average basket" description is insufficient. The result supports monitoring small and large shops separately, but size alone is not treated as a customer segment.

3.3 PCA Variance Structure

Twenty PCs retain 54.92% of total feature variance; the first two retain only 11.19%. No small set of components captures nearly all variation. This is consistent with heterogeneous basket compositions rather than one dominant low-dimensional process.

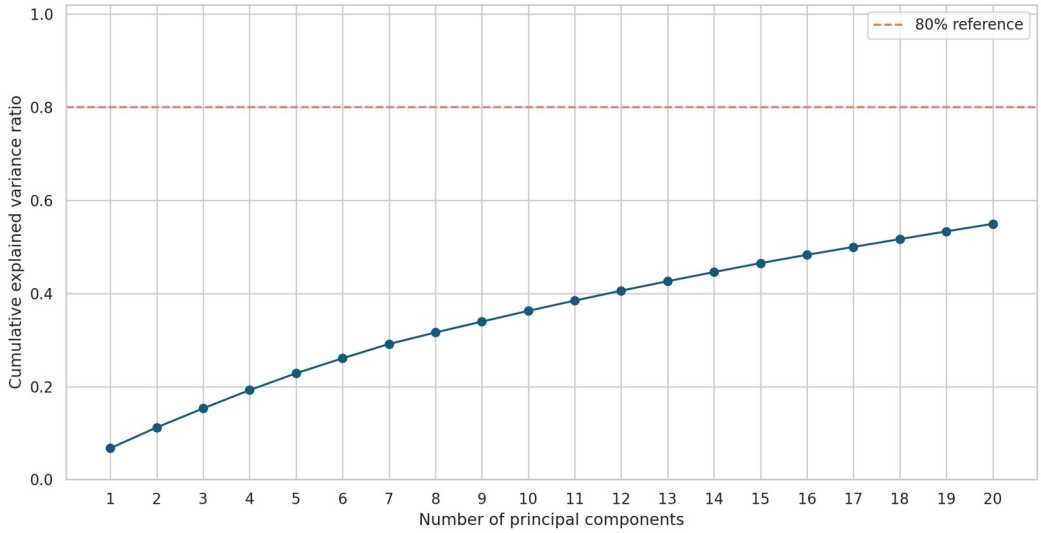


Figure 1.3: Cumulative explained variance for the first 20 principal components

Interpretation. The gradual curve means that 20-component clustering retains useful information but discards 45.08% of variance. Two-dimensional PCA plots are only visual windows, not complete representations of cluster separation.

4. MODEL EVALUATION

4.1 K Selection

K = 2 has the highest sampled silhouette (0.1832), while K = 5 has silhouette 0.1597 and DBI 2.0806. K = 5 improves DBI relative to K = 2–4 and yields more detailed basket profiles. K = 7 has a slightly lower DBI (1.9696), but additional fragmentation was not justified by the operational objective. K = 5 is therefore retained as a deliberate trade-off between compactness and interpretability.

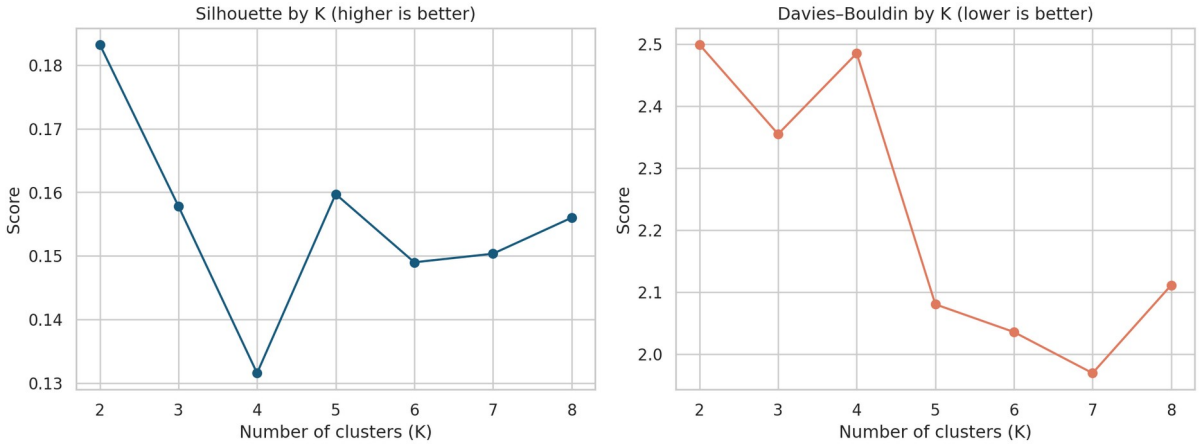


Figure 1.4: K-means silhouette and Davies–Bouldin diagnostics for K = 2,…,8

Interpretation. The two criteria do not identify the same optimum. Selecting solely by silhouette would produce a coarse two-group split; selecting solely by DBI would favour seven groups. The modest absolute silhouette values indicate overlapping structure throughout the scan.

4.2 Comparative Performance

Table 1.3: Internal validation on the fixed 2,000-transaction evaluation sample

Model	Silhouette (higher is better)	DBI (lower is better)	Coverage	Evaluation condition
K-means	0.1605	2.0539	100%	Full-data fit; five clusters
Ward hierarchical	0.0563	3.2505	100%	Full-data fit; five clusters
DBSCAN	0.2231	1.0422	81.55%	Core points only; standardized PCs
Spectral	−0.1207	2.1623	100%	Sample fit; nearest-neighbour graph

DBSCAN achieves the highest core-only silhouette and lowest DBI, but it labels **1,317 of 7,500 full-data transactions (17.56%) as noise**. Its score is not a free victory: roughly one in six evaluation-sample transactions does not contribute to those metrics. K-means offers weaker but positive geometric separation with complete coverage. Ward performs poorly on both criteria. Spectral produces a negative silhouette and highly unequal groups, including very small clusters, so the tested graph configuration is unsuitable as the primary taxonomy.

Decision. Use K-means for a complete five-way basket taxonomy and DBSCAN as a complementary atypical-basket detector. Do not deploy spectral clustering in the tested configuration, and do not treat Ward's hierarchy as evidence of stronger business segments.

4.3 Hierarchical Structure

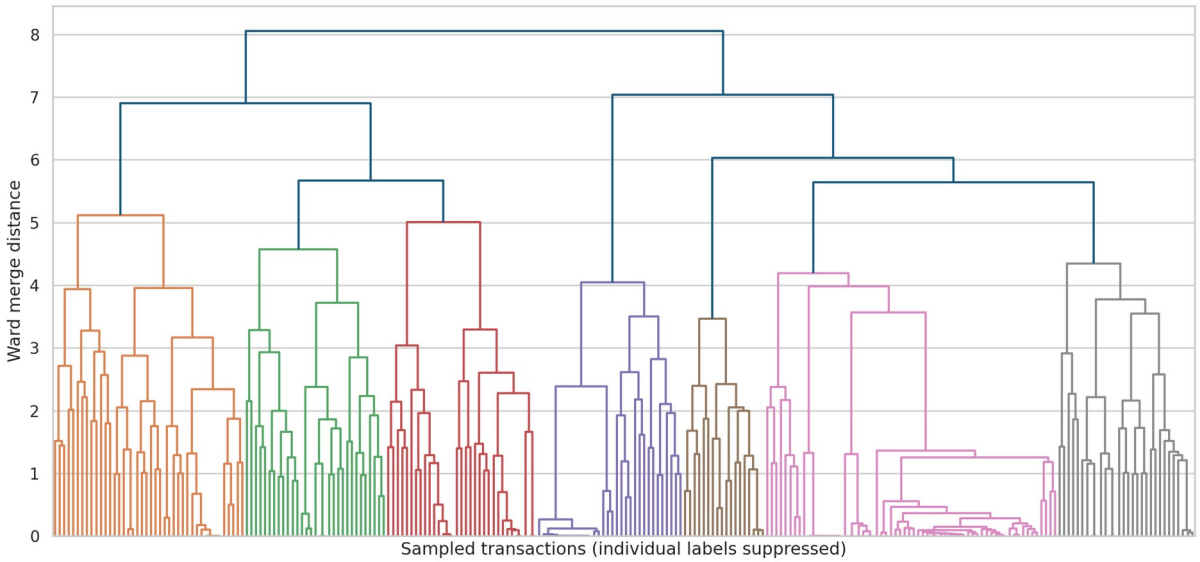


Figure 1.5: Ward dendrogram for a readable random sample of 250 baskets

Interpretation. Increasing merge distances reveal nested dissimilarity, but there is no single visually unambiguous cut that proves five clusters. The dendrogram supports hierarchical exploration, while the weak full-model scores caution against overinterpreting its branches.

4.4 K-means Geometry and Basket Profiles

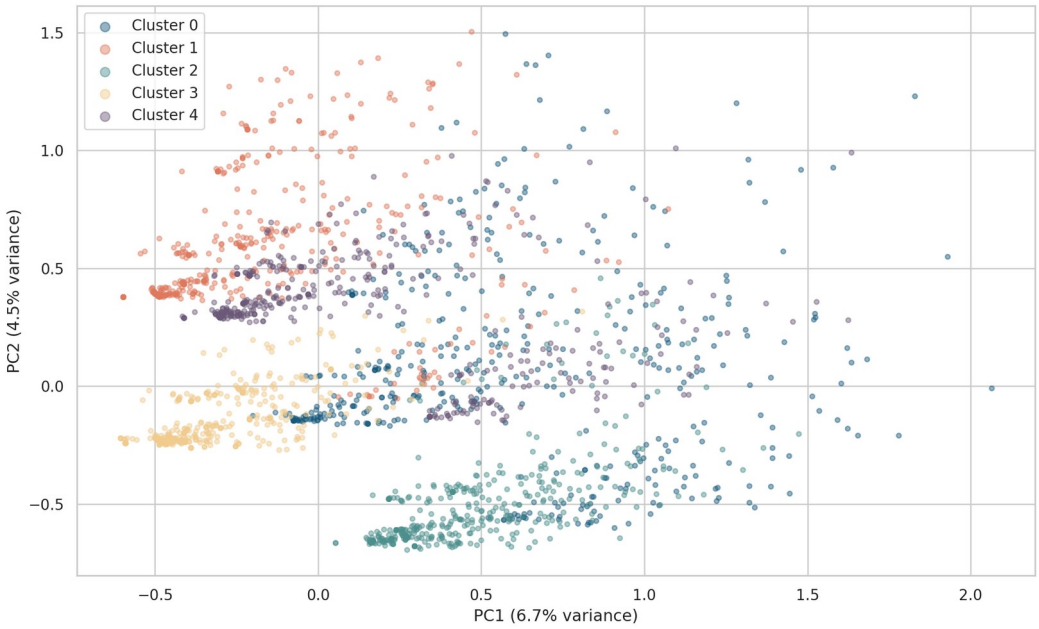


Figure 1.6: K-means assignments displayed in the first two PCA dimensions

Interpretation. Elongated bands and overlaps remain visible. Apparent visual gaps are not conclusive because the plot retains only 11.19% of original variance, while fitting uses 20 PCs. The modest silhouette is consistent with partial overlap.

Table 1.4: Interpretable K-means basket groups

Group	Baskets	Mean size	Anchor and supporting products	Provisional label
C0	1,243	6.39	Spaghetti 99.9%; mineral water 36.0%; ground beef 23.7%	Spaghetti-centred larger baskets
C1	1,081	4.40	French fries 100%; eggs 20.3%; chocolate 17.8%	Fries-centred mixed baskets
C2	964	4.62	Mineral water 100%; chocolate 19.7%; milk 17.8%	Water-centred top-up baskets
C3	3,341	2.50	Chocolate 13.0%; green tea 12.2%; cookies 11.7%	Small mixed baskets
C4	871	4.41	Eggs 100%; mineral water 29.4%; milk 15.7%	Egg-centred mixed baskets

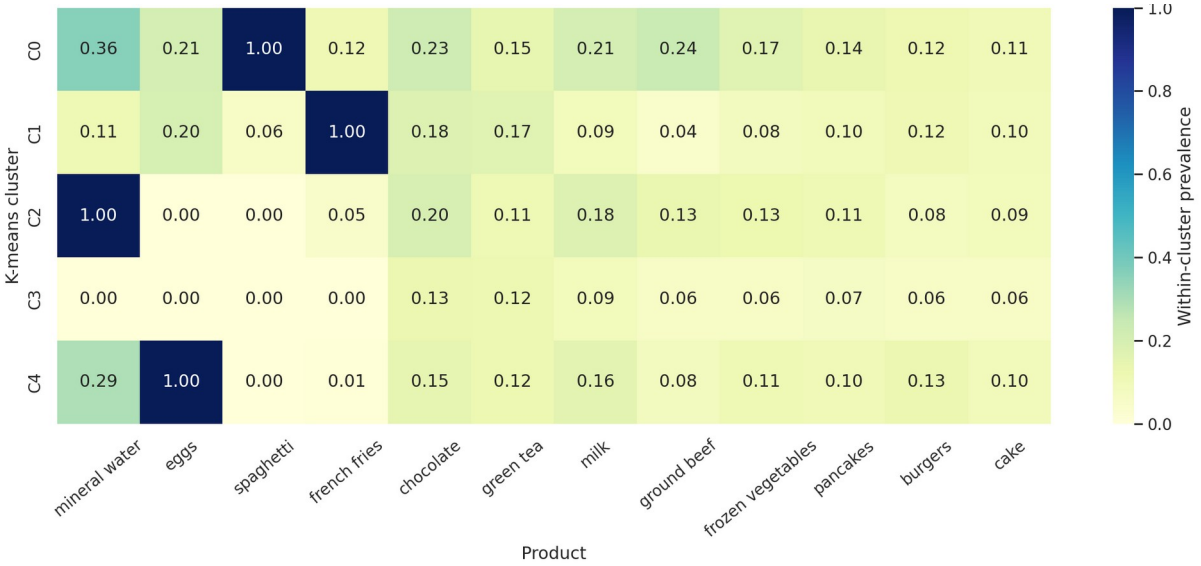


Figure 1.7: Within-cluster prevalence of the 12 most frequent products

Interpretation. Four groups are almost deterministically defined by one anchor item; C3 instead captures small mixed transactions. This is easy to operationalize, but such sharp boundaries may reflect the dataset's generating structure. They require temporal/out-of-sample validation before being called genuine shopping missions.

4.5 Limitations and Validity Threats

- **Transaction rather than customer inference:** repeated rows cannot be connected to individuals, so loyalty or persona claims are unsupported.
- **Binary and static data:** quantities, value, seasonality, and sequence are absent; high frequency is not high profitability.
- **Representation sensitivity:** Euclidean PCA distances need not match Jaccard similarity for sparse sets. The 20-PC solution omits 45.08% of variance.
- **Algorithm sensitivity:** K, DBSCAN radius, graph neighbours, and scaling alter outcomes. Spectral sampling and DBSCAN noise removal limit direct comparability.
- **Internal-index scope:** silhouette and DBI favour compact geometric partitions and do not measure incremental sales or business usefulness.

- **Anchor-product risk:** near-100% prevalence can produce convincing-looking but trivial rules. Clusters should be tested on later transactions and alternative stores.
- **No external ground truth:** there is no independently labelled mission variable against which to validate semantic accuracy.

5. CONCLUSION

5.1 Conclusion

The analysis converts an unconventional positional transaction file into a reproducible basket-level workflow. The store's transactions are predominantly small: the median basket has three products, 41.49% contain at most two, and only 5.25% contain ten or more. A small set of products—especially mineral water, eggs, spaghetti, and french fries—has high support and strongly shapes subsequent clustering.

No algorithm dominates every criterion. K-means at K = 5 provides the most practical complete taxonomy, with five product-interpretable basket groups and full coverage. DBSCAN has the strongest core-only silhouette and DBI, but excludes 17.56% of transactions as noise; it is more appropriate as a secondary atypical-basket detector than as a replacement taxonomy. Ward is weak under both internal indices, and the tested spectral solution has negative silhouette and unstable balance.

The defensible conclusion is therefore measured: clustering reveals useful, interpretable regularities in transaction composition, but the present evidence does not prove stable customer segments or financial benefit. The most valuable next step is out-of-sample and experimental validation, not further optimization of an internal index.

5.2 Business Implications

- **Recommendations:** use anchor-product profiles to design candidate complementary-product recommendations, but run controlled experiments before claiming uplift.
- **Assortment review:** inspect DBSCAN-noise baskets for unusual combinations, emerging demand, or recording problems; do not automatically discard them.
- **Monitoring:** track basket-size distribution, cluster mix, and product support on new periods. Material drift should trigger refitting and profile review.
- **Commercial validation:** join future data to quantity, revenue, margin, timestamp, and privacy-compliant customer identifiers. Evaluate recommendations using incremental margin rather than cluster scores.

5.3 Overall Conclusion

This case study analysed 7,500 retail transactions end to end: data audit and cleaning, descriptive statistics, binary basket vectors, PCA, four clustering algorithms, and internal validation. The data are dominated by small baskets (median of three products, 41.49% with at most two) and a few high-support staples, which strongly shape every clustering result.

Among the algorithms, K-means with K = 5 gave the most useful complete taxonomy (silhouette 0.1605, DBI 2.0539, 100% coverage), organised around the anchor products spaghetti, mineral water, eggs, and french fries plus a large small-basket group. DBSCAN scored best on the core points it kept (silhouette 0.2231, DBI 1.0422) but left 17.56% of transactions as noise, so it is best used to flag atypical baskets. Ward hierarchical and the tested spectral configuration were not competitive. Overall cluster separation is weak, and the groups should be treated as hypotheses to validate on new data, not as proven customer segments.

CASE STUDY 4

"Predicting Student Dropout and Academic Success Using Data Science"

Fundamentals of Data Science (22CSC59)

Sriya Ammanamanchi (160124733156) | Section: CSE-3 | Year: 2026-27

Case Study at a Glance

Dataset	4,424 undergraduate students, 36 predictors and a three-class target (Dropout / Enrolled / Graduate)
Techniques	Exploratory data analysis, Kruskal-Wallis hypothesis test with effect size, stratified split, encoding and scaling, multinomial logistic regression
Evaluation	Confusion matrix, per-class and macro-averaged precision, recall and F1-score, effect of class imbalance
Headline result	Accuracy 75.0% and macro F1-score 0.669 from first-semester information; first-semester grades differ significantly across outcomes ($H = 1094.19$, $\varepsilon^2 = 0.247$)

1. INTRODUCTION

1.1 Problem Definition, Significance and Objectives

Problem Definition

Higher-education institutions lose a substantial share of their students before the course is completed. A student who leaves represents lost opportunity for the individual and lost investment for the institution, and in many cases the warning signs are already present in the student's records long before the exit happens. These records include the student's age and background, how the student entered the programme, whether the tuition fees are being paid, and how the student performed in the first semester.

This case study works with a real dataset of **4,424 undergraduate students** from a Portuguese higher-education institution. Each student is labelled with the status reached at the end of the normal course duration: **Dropout**, **Enrolled** (still studying) or **Graduate**. The task is to study how the available attributes relate to these outcomes, test one of those relationships formally, and build a multiclass classification model that predicts the outcome. Because the label describes the end of the course, the study also has to decide carefully which attributes would genuinely be known early enough to support an early prediction.

Significance

If the factors linked to dropout are understood, support staff can concentrate counselling, tutoring and financial assistance on students who need them while there is still time to help. A predictive model is useful only as a screening aid for this purpose. It must not be used to label or penalise students, and the statistical associations it finds must not be mistaken for causes. These cautions shape the way results are interpreted throughout the report.

Objectives

The objectives of this case study are:

- To load, inspect and clean the student dataset, and to classify each attribute by what its encoded values actually mean.
- To explore how admission grade, first-semester academic performance, age at enrolment and financial status relate to the final outcome.
- To test formally whether first-semester grades differ across the three outcome groups, using a test whose assumptions are checked.
- To build a multiclass logistic regression model using only information available early, and to evaluate it using per-class and macro-averaged metrics rather than accuracy alone.
- To interpret the model, state its limitations, and recommend practical and responsible support measures for at-risk students.

1.2 Concepts Used in Methodology

The following ideas from statistics and machine learning are used in this study.

- **Exploratory Data Analysis (EDA):** summarising and visualising the data with counts, descriptive statistics, histograms, box plots and stacked bar charts to reveal distributions and relationships before any modelling.
- **Data types by meaning:** a column stored as an integer is not necessarily a number. Codes such as *Course* or *Application mode* are nominal categories and must be treated as such.

- **Hypothesis testing:** a formal procedure for deciding whether an observed difference could plausibly be due to chance. Here the Kruskal-Wallis H test, a rank-based test for three or more independent groups, is used because the normality and equal-variance assumptions of ANOVA fail.
- **Effect size:** a measure of how large a difference is, as opposed to whether it is statistically detectable. Epsilon-squared is reported for the Kruskal-Wallis test.
- **Stratified train-test split:** dividing the data so that the class proportions are the same in the training and testing parts.
- **Encoding and scaling:** one-hot encoding turns nominal codes into 0/1 indicator columns, and z-score scaling places numerical variables on a common scale.
- **Multinomial logistic regression:** a linear model that estimates the probability of each of the three classes through the softmax function, with L2 regularisation to keep the coefficients stable.
- **Evaluation metrics:** the confusion matrix, accuracy, precision, recall and F1-score, together with macro averages that give every class equal weight.
- **Class imbalance and data leakage:** the outcome classes are of unequal size, and some columns (second-semester results) would not be known at the time of an early prediction. Both issues are handled explicitly.

1.3 Outline of the Results

Half of the students (49.9%) graduated, 32.1% dropped out and 17.9% were still enrolled. First-semester academic performance, tuition and debt status, scholarship status and age at enrolment show the clearest relationships with the outcome, whereas admission grade differs only slightly between the groups. The Kruskal-Wallis test shows that first-semester grades differ significantly across the three outcomes ($H = 1094.19$, $p < 0.001$) with a large effect size ($\varepsilon^2 = 0.247$).

A multinomial logistic regression built from enrolment and first-semester information reaches an accuracy of 75.0% and a macro-averaged F1-score of 0.669 on 885 unseen students. It identifies Graduate and Dropout students well but is weak for the Enrolled class (F1 = 0.396). Adding second-semester data improves the macro F1-score only to 0.695, which supports the use of the earlier, more useful model. Recommendations and conditions for responsible use are given in Chapter 7.

1.4 Tools and Environment

All work was carried out in **Python 3** using a **Jupyter Notebook**. The libraries used are pandas and NumPy for data handling, SciPy for statistical tests, scikit-learn for preprocessing, modelling and evaluation, and matplotlib and seaborn for visualisation. The executable notebook (*Student_Dropout_Case_Study.ipynb*) accompanies this report and reproduces every number, table and figure shown here.

2. DATASET DESCRIPTION AND PREPROCESSING

This chapter covers Part A of the case study: loading the data, understanding its structure, categorising the attributes, checking quality, examining distributions and deciding which preprocessing steps are required.

2.1 Loading the Data

The file *Studentdata.csv* uses a semicolon (;) as separator, so it must be read with `sep=";"` in pandas; otherwise the entire table collapses into a single column. After loading, two small corrections were made to the column names: a hidden tab character inside *Daytime/evening attendance* was removed, and the misspelt *Nacionality* was renamed *Nationality*.

Table 4.1: Overview of the dataset

Property	Value
Rows (students)	4,424
Columns	37 (36 predictors + 1 target)
Data types	29 integer, 7 float, 1 text (Target)
Target variable	Target: Dropout / Enrolled / Graduate
Missing values	0
Duplicate rows	0

The first five records are shown in Table 4.2 (a selection of columns is shown for readability; the full 37-column view is in the notebook). The first student is a Dropout who had no approved units and no graded units in the first semester, while the second and fourth are Graduates with a first-semester grade above 13.

Table 4.2: First five records (selected columns)

Marital status	App. mode	Course	Adm. grade	Age	Sem-1 approved	Sem-1 grade	Target
1	17	171	127.3	20	0	0.00	Dropout
1	15	9254	142.5	19	6	14.00	Graduate
1	1	9070	124.8	19	0	0.00	Dropout
1	17	9773	119.6	20	6	13.43	Graduate
2	39	8014	141.5	45	5	12.33	Graduate

Most columns are stored as integers, but the stored type is misleading. For example, *Course = 9254* is an identifier and *Marital status = 1* means single. The treatment of each variable therefore has to be decided from its meaning and not from its data type.

2.2 Target Variable and Predictors

The target variable is **Target**, with three categories: Dropout, Enrolled and Graduate. The remaining 36 columns are potential predictors. Of these, six describe second-semester results (credited, enrolled, evaluations, approved, grade and without evaluations). Since the label describes the end of the course, using second-semester results would give the model information that arrives late, so they are excluded from the main model and used only in a comparison (Section 6.4).

2.3 Categorisation of Attributes

Table 4.3: Classification of attributes by meaning

Type	Attributes	Reasoning
Numerical (18)	Age at enrollment, Admission grade, Previous qualification (grade), all Curricular units columns for semester 1 and 2 (credited, enrolled, evaluations, approved, grade, without evaluations), Unemployment rate, Inflation rate, GDP	True quantities; arithmetic is meaningful
Binary (8)	Daytime/evening attendance, Displaced, Educational special needs, Debtor, Tuition fees up to date, Gender, Scholarship holder, International	Only values 0 and 1
Nominal categorical (9)	Marital status, Application mode, Course, Previous qualification, Nationality, Mother's and Father's qualification, Mother's and Father's occupation	Integers are labels with no order
Ordinal (1)	Application order (0 = first choice to 9 = last choice)	Order matters; spacing does not

2.4 Missing Values, Duplicates and Unusual Values

The dataset has **no missing values and no duplicate rows**, so neither imputation nor de-duplication is required. All eight binary columns were verified to contain only 0 and 1. Several unusual patterns were nevertheless found:

- **Zero first-semester grades:** 718 students (16.2%) have a first-semester grade of exactly 0, and 180 of them were enrolled in zero units. A zero here means that no graded work was recorded, not that the student scored zero. These records are meaningful and are retained.
- **Zero grades by outcome:** 40.1% of Dropouts, 8.9% of Enrolled and 3.5% of Graduates have a zero first-semester grade, which already points to a strong link with the outcome.
- **Age:** the ages range from 17 to 70, with 43 students older than 50. These are genuine mature students and are retained.
- **Consistency check:** no student has more approved units than enrolled units in semester 1, so the unit counts are internally consistent.

2.5 Distributions of Selected Numerical Attributes

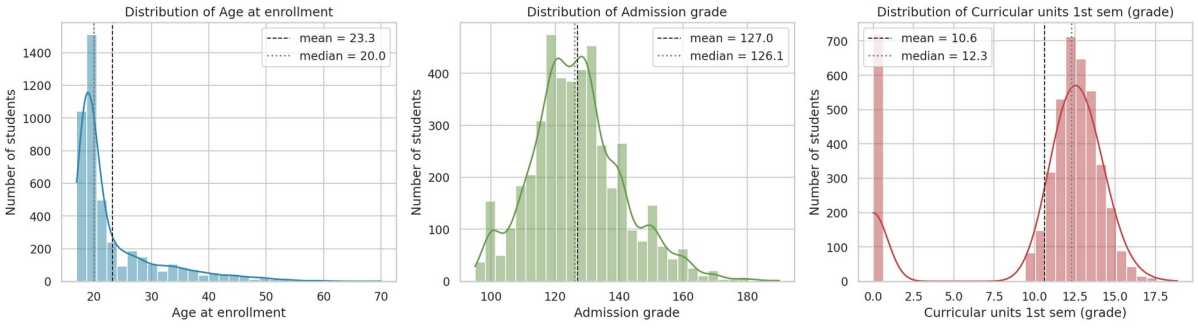


Figure 4.1: Histograms of age at enrolment, admission grade and first-semester grade

Interpretation. Age at enrolment is strongly right-skewed (skewness 2.05): most students begin at 18 to 20, so the median (20) is lower than the mean (23.3), which is pulled upward by a tail of mature students reaching 70. Admission grade is approximately bell-shaped, centred near 127 on a 0 to 200 scale, with mild right skew (0.53). First-semester grade is left-skewed (−1.57) with a distinct spike at 0 corresponding to students with no graded units, followed by a main cluster between about 11 and 14. The spike makes the variable clearly non-normal, which influences the choice of statistical test in Chapter 4.

2.6 Preprocessing Required Before Modelling

- **Cleaning:** column names were tidied; no records need to be removed or imputed.
- **One-hot encoding of nominal codes** (Course, Application mode, qualifications, occupations, marital status, nationality). Feeding the raw codes to a linear model would imply a false numerical order. Categories with fewer than 20 students are grouped together to avoid unstable coefficients.
- **Standardisation (z-score) of numerical variables and Application order,** because logistic regression with regularisation is sensitive to variable scale.
- **Binary variables** are left as 0/1.
- **Avoiding leakage:** second-semester columns are excluded from the main model, and the encoder and scaler are fitted only on the training data through a scikit-learn Pipeline.
- **Stratified splitting** is used to preserve the class proportions, since the classes are imbalanced.

3. EXPLORATORY DATA ANALYSIS AND STATISTICS

This chapter covers Part B. It describes the outcome distribution and examines how admission grade, first-semester performance, age and financial indicators relate to the outcome.

3.1 Distribution of Student Outcomes

Table 4.4: Number and percentage of students in each outcome

Outcome	Count	Percentage
Graduate	2,209	49.93%
Dropout	1,421	32.12%
Enrolled	794	17.95%
Total	4,424	100.00%

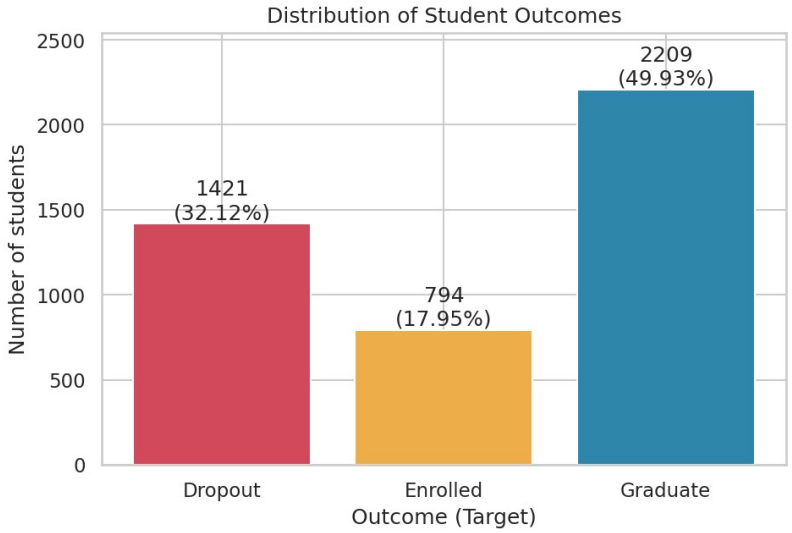


Figure 4.2: Distribution of student outcomes (Dropout, Enrolled, Graduate)

Interpretation. Graduates form the largest group, followed by Dropouts, and Enrolled students are the smallest class at under 18%. The Graduate class is nearly three times larger than the Enrolled class, so the data are **imbalanced**. A model that favours the larger classes could achieve a respectable accuracy while performing poorly on the Enrolled group, which is why macro-averaged metrics are used in the evaluation.

3.2 Descriptive Statistics

Table 4.5: Descriptive statistics of selected numerical variables (all students)

Variable	Mean	Std	Min	25%	Median	75%	Max
Age at enrollment	23.27	7.59	17	19	20	25	70
Admission grade	126.98	14.48	95.0	117.9	126.1	134.8	190.0
Sem-1 units approved	4.71	3.09	0	3	5	6	26
Sem-1 grade	10.64	4.84	0	11.0	12.29	13.4	18.88

Table 4.6: Mean of selected variables by outcome

Variable (mean)	Dropout	Enrolled	Graduate
Age at enrollment	26.07	22.37	21.78
Admission grade	124.96	125.53	128.79
Sem-1 units approved	2.55	4.32	6.23
Sem-1 grade	7.26	11.13	12.64

Interpretation. The average student is 23.3 years old, with an admission grade of 127 and about 4.7 approved units in the first semester. The group means in Table 4.6 show the main pattern of the study: the average number of approved first-semester units is 6.23 for Graduates, 4.32 for Enrolled students and only 2.55 for Dropouts. Dropouts are also older on average (26.1 years against 21.8 for Graduates), while differences in admission grade between the groups are small.

3.3 Admission Grade Across Outcomes

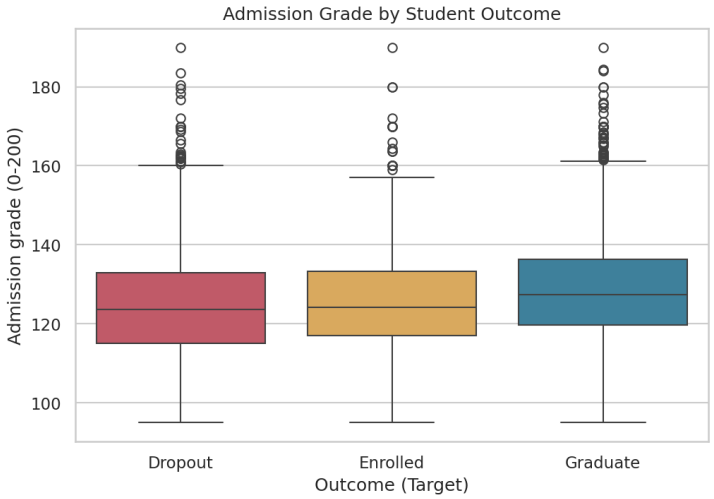


Figure 4.3: Box plot of admission grade by student outcome

Interpretation. The three boxes overlap substantially. The median admission grades are 123.6 for Dropouts, 124.1 for Enrolled students and 127.4 for Graduates. A Kruskal-Wallis test confirms that the difference is statistically significant ($H = 73.23$, $p = 1.25 \times 10^{-16}$), but the gap is only about 3 to 4 points on a 200-point scale. Admission grade therefore separates the outcomes only weakly; its significance is largely a consequence of the large sample size.

3.4 First-Semester Academic Performance and Outcome

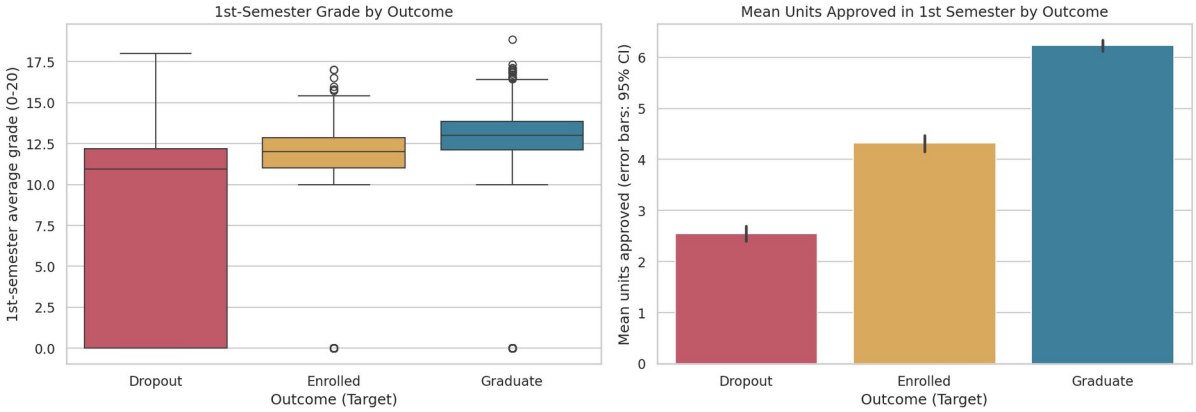


Figure 4.4: First-semester grade (left) and mean approved units (right) by outcome

Table 4.7: First-semester grade by outcome

Outcome	Count	Mean	Std	25%	Median	75%
Dropout	1,421	7.26	6.03	0.00	10.93	12.20
Enrolled	794	11.13	3.68	11.00	12.00	12.86
Graduate	2,209	12.64	2.70	12.12	13.00	13.86

Interpretation. First-semester performance separates the groups much more clearly than admission grade. The median first-semester grade is 13.0 for Graduates, 12.0 for Enrolled students and 10.9 for Dropouts. The Dropout box extends all the way down to 0 because 40.1% of Dropouts have no graded units, and the Dropout standard deviation (6.03) is more than double that of Graduates (2.70). The right-hand panel shows that the mean number of approved units falls from 6.2 to 4.3 to 2.6 as we move from Graduates to Enrolled to Dropouts, and the 95% confidence intervals do not overlap. Early academic engagement is therefore a strong warning signal.

3.5 Age at Enrolment and Outcome

Table 4.8: Outcome percentage within each age group

Age group (years)	Dropout %	Enrolled %	Graduate %
19 or under	21.0	17.0	62.1
20 to 24	28.6	22.6	48.9
25 to 29	57.6	16.9	25.5
30 and above	54.1	12.7	33.2

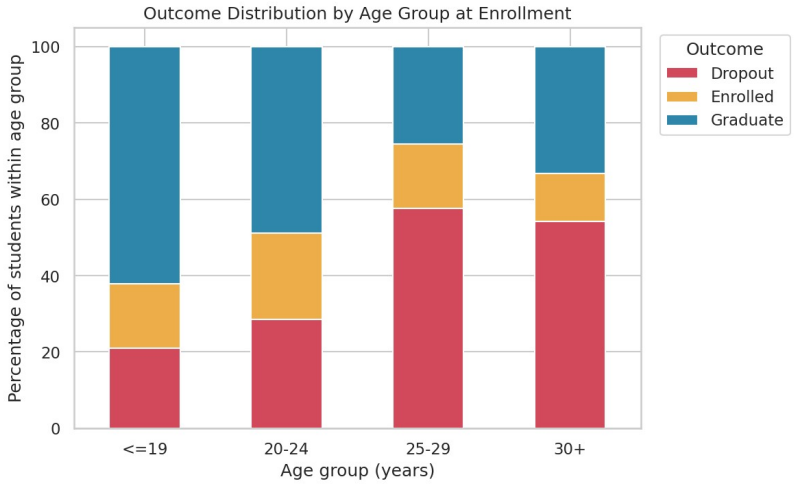


Figure 4.5: Outcome distribution by age group at enrolment

Interpretation. The share of Dropouts rises from 21.0% among students aged 19 or under to 28.6% for ages 20 to 24, then jumps to 54% to 58% for students aged 25 and above. The change occurs at the threshold of 25 years and is not gradual. A chi-square test of independence confirms that age group and outcome are associated ($\chi^2 = 452.0$, df = 6, $p < 0.001$). Work and family commitments are plausible explanations but are not measured in this dataset, so the finding is an association and not a demonstrated cause.

3.6 Financial Indicators and Outcome

Table 4.9: Dropout rate by financial indicators

Indicator	Value	Dropout rate (%)	Students (n)
Tuition fees up to date	No	86.6	528
Tuition fees up to date	Yes	24.7	3,896
Debtor	No	28.3	3,921
Debtor	Yes	62.0	503
Scholarship holder	No	38.7	3,325
Scholarship holder	Yes	12.2	1,099

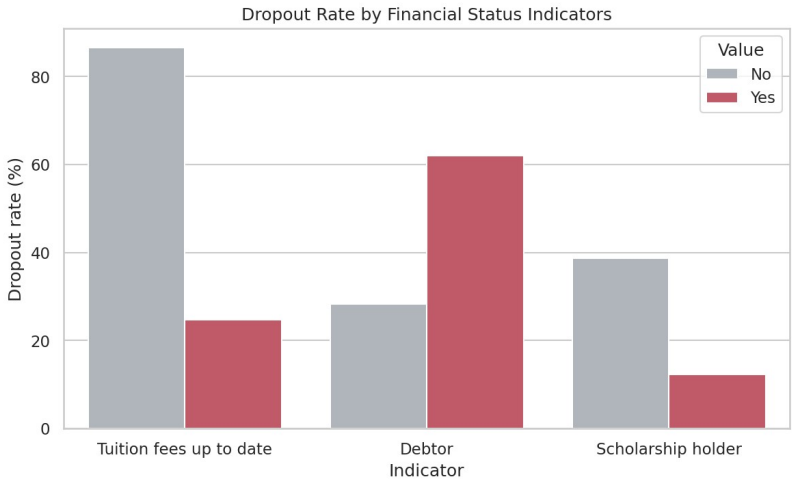


Figure 4.6: Dropout rate by tuition, debt and scholarship status

Interpretation. Financial status shows the sharpest contrasts in the whole dataset. Among the 528 students whose fees are not up to date, 86.6% dropped out, compared with 24.7% of students whose fees are paid. Debtors drop out at 62.0% against 28.3% for non-debtors, while scholarship holders drop out at only 12.2% against 38.7%. Financial strain is clearly linked to leaving, although the data cannot show whether it causes students to leave, results from them disengaging, or both.

3.7 Summary of Exploratory Findings

- The outcome classes are imbalanced (49.9% / 32.1% / 17.9%).
- First-semester units approved and grades differ strongly across outcomes; admission grade differs only slightly.
- Dropout is much more common from the age of 25 upward.
- Unpaid fees and debt are associated with very high dropout rates, and scholarships with low dropout rates.

4. HYPOTHESIS TESTING

This chapter covers Part C. The research question is whether first-semester academic performance is related to the final outcome.

4.1 Research Question and Hypotheses

Do first-semester grades (average grade of curricular units) differ across the three outcome groups, Dropout, Enrolled and Graduate?

- **Null hypothesis (H_0):** the distribution of first-semester grades is the same in all three outcome groups.
- **Alternative hypothesis (H_1):** the distribution of first-semester grades differs in at least one outcome group.
- **Significance level:** $\alpha = 0.05$.

4.2 Selection of the Test and Assumption Checks

The comparison involves one numerical variable and three independent groups, for which the classical test is one-way ANOVA. ANOVA assumes independent observations, approximately normal data within each group, and equal variances across groups. The second and third assumptions were checked before the test was selected.

Table 4.10: Assumption checks for one-way ANOVA

Check	Result	Conclusion
Independence	Each row is a different student	Reasonable
Normality (Shapiro-Wilk, sample up to 1,000 per group)	W = 0.745 (Dropout), 0.629 (Enrolled), 0.615 (Graduate); all $p < 0.001$; group skewness −0.31, −2.36, −3.45	Violated
Equal variances (Levene)	W = 549.9, $p < 0.001$; standard deviations 6.03, 3.68, 2.70	Violated

Both assumptions fail clearly. With 4,424 observations ANOVA is fairly tolerant of mild departures, but the spike of zero grades and the very different spreads are not mild. The **Kruskal-Wallis H test** was therefore chosen as the primary test. It compares the rank distributions of the groups and needs neither normality nor equal variances. ANOVA is reported only as a sensitivity check.

4.3 Test Results

Table 4.11: Results of the hypothesis tests on first-semester grades

Quantity	Value
Kruskal-Wallis H statistic (df = 2)	1094.19
p-value	2.51×10^{-238}
Effect size $\varepsilon^2 = (H - k + 1)/(n - k)$	0.247 (large)
ANOVA sensitivity check	$F = 713.5$, $p = 2.80 \times 10^{-269}$
Post-hoc Dropout vs Enrolled (Mann-Whitney U)	U = 368,444; $p = 6.9 \times 10^{-43}$
Post-hoc Dropout vs Graduate	U = 600,098; $p = 2.0 \times 10^{-218}$

Quantity	Value
Post-hoc Enrolled vs Graduate	$U = 524{,}677$; $p = 1.9 \times 10^{-63}$
Excluding grade = 0 (medians 12.0 / 12.1 / 13.0)	$H = 474.5$; $p = 9.3 \times 10^{-104}$

Decision. Since $p < 0.05$, the null hypothesis is rejected: first-semester grades differ significantly among the three outcome groups. The pairwise Mann-Whitney tests, judged against a Bonferroni-adjusted level of $0.05/3 = 0.0167$, show that all three pairs differ. The ANOVA result agrees with the Kruskal-Wallis conclusion.

4.4 Statistical Significance versus Practical Importance

With more than four thousand students, even a trivial difference can produce a very small p-value, so the p-value alone is not informative about importance. The effect size $\varepsilon^2 = 0.247$ indicates that roughly a quarter of the variation in grade ranks is associated with outcome group, which is a large effect.

The size of the difference needs care, however. The median grade differs by about 2 points between Graduates (13.0) and Dropouts (10.9), but the mean gap is larger (12.6 against 7.3) because 40% of Dropouts have a grade of 0. When students with a zero grade are excluded, the medians become 12.0, 12.1 and 13.0, the difference remains statistically significant, but it shrinks to roughly one grade point. The result is therefore partly an engagement effect (whether the student took any assessed units at all) and partly a modest genuine difference in grade level. It is both statistically significant and practically meaningful, but it describes association and not causation.

5. CLASSIFICATION MODEL

This chapter covers the model-building steps of Part D. Evaluation of the model is presented separately in Chapter 6.

5.1 Separating Predictors from the Target

The target is `Target`. For the main model, the 36 candidate predictors were reduced to **30** by excluding the six second-semester columns. This design choice turns the problem into an *early-warning* task: the model uses only what is known at enrolment (demographic, socio-economic and admission information, and macroeconomic indicators) plus first-semester academic results. A second model that includes semester-2 data was also trained, purely as a comparison.

5.2 Train-Test Split

The data were split 80:20 using stratified sampling with a fixed random seed (42). Stratification keeps the class proportions identical in both parts, which matters because the Enrolled class is small and an unstratified split could leave the test set with too few Enrolled students for reliable evaluation.

Table 4.12: Train-test split

Part	Students	Dropout / Enrolled / Graduate (%)	Purpose
Training	3,539	32.1 / 17.9 / 49.9	Fit the model
Testing	885	32.1 / 18.0 / 49.9	Final evaluation (unseen)

5.3 Encoding and Feature Scaling

A scikit-learn `ColumnTransformer` was used to apply different treatments to different attribute types: z-score scaling for numerical variables and Application order, one-hot encoding for the nominal codes (categories with fewer than 20 students grouped, and unseen categories ignored), and pass-through for binary variables. The transformer and the classifier were combined into a single `Pipeline`, so that the scaler and encoder are fitted on the training data only. This prevents information from the test set leaking into the model.

5.4 Multiclass Logistic Regression

The classifier is scikit-learn's `LogisticRegression`, which fits a multinomial (softmax) model for the three classes, with L2 regularisation (C = 1) and up to 3,000 iterations to ensure convergence. Predictions were generated for the 885 test students. As a check on overfitting, 5-fold stratified cross-validation was also run on the training set; it gave a macro F1-score of 0.665 ± 0.012, which agrees closely with the test result reported next.

6. MODEL EVALUATION

6.1 Confusion Matrix

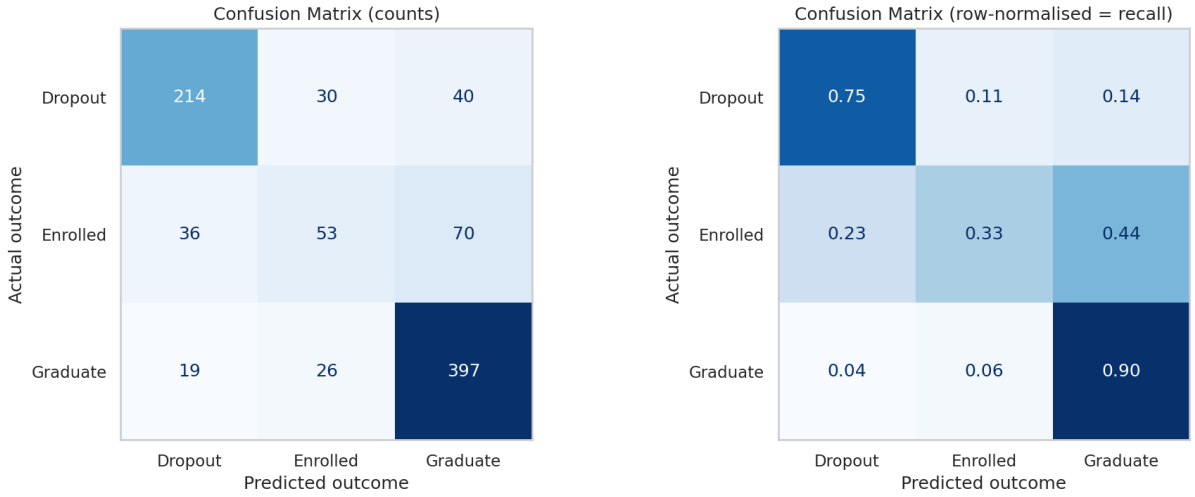


Figure 4.7: Confusion matrix of the early model on the test set (counts and row-normalised)

Table 4.13: Confusion matrix (counts)

	Predicted Dropout	**Predicted Enrolled**	**Predicted Graduate**
Actual Dropout	214	30	40
Actual Enrolled	36	53	70
Actual Graduate	19	26	397

Interpretation. The diagonal shows correct predictions: 214 of 284 Dropouts (75%) and 397 of 442 Graduates (90%) were classified correctly. The Enrolled class fares much worse: only 53 of 159 Enrolled students (33%) were identified. 70 were predicted to graduate and 36 to drop out. The row-normalised panel makes the pattern clear, because the Enrolled row has no strong diagonal.

6.2 Performance Metrics

Table 4.14: Classification performance of the early model (test set, n = 885)

Class	**Precision**	**Recall**	**F1-score**	**Support**
Dropout	0.796	0.754	0.774	284
Enrolled	0.486	0.333	0.396	159
Graduate	0.783	0.898	0.837	442
Macro average	0.688	0.662	0.669	885
Weighted average	0.734	0.750	0.737	885

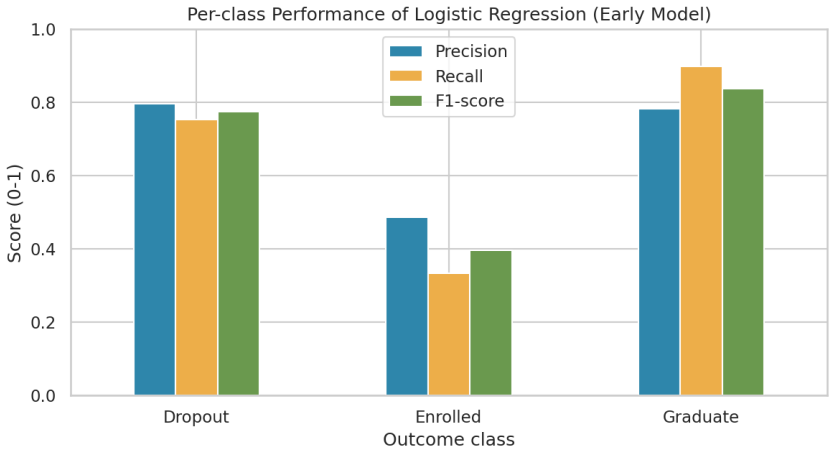


Figure 4.8: Per-class precision, recall and F1-score of the early model

Interpretation. Overall accuracy is 75.0%, but this single number hides large differences between the classes. For **Graduate**, precision is 0.783 and recall 0.898 (F1 = 0.837). For **Dropout**, precision is 0.796 and recall 0.754 (F1 = 0.774), meaning that the model finds three-quarters of actual dropouts and about four out of five students it flags as likely dropouts really do leave. For **Enrolled**, precision is only 0.486 and recall 0.333 (F1 = 0.396).

The macro-averaged scores (precision 0.688, recall 0.662, F1 0.669) are lower than accuracy because every class is given equal weight and the weak Enrolled class pulls the average down. The weighted F1 of 0.737 is higher because it is dominated by the larger classes. The Enrolled students are an in-between group: their first-semester profile lies between those of Dropouts and Graduates, so a linear model cannot separate them cleanly. The cross-validated macro F1 of 0.665 agrees with the test value, so the model is not overfitted.

6.3 Does the Model Perform Differently Across Classes?

Yes. The F1-scores of 0.837 (Graduate), 0.774 (Dropout) and 0.396 (Enrolled) show that performance is unequal. The model is reliable enough to support screening for likely dropouts and likely graduates, but its predictions for Enrolled students should not be relied on.

6.4 Cost of Predicting Early

Table 4.15: Early model versus model with semester-2 data (test set)

Metric	Early model (semester 1)	Full model (semesters 1 and 2)
Accuracy	0.750	0.768
Macro precision	0.688	0.712
Macro recall	0.662	0.687
Macro F1-score	0.669	0.695
Enrolled F1-score	0.396	0.457

Interpretation. Adding second-semester results increases accuracy by only 1.8 percentage points and the macro F1-score by 0.026. The early model therefore keeps most of the predictive power and gives the institution a full extra semester of lead time, which justifies the early-prediction design.

6.5 Effect of Class Imbalance

To study the imbalance, the model was refitted with `class_weight="balanced"`, which penalises mistakes on small classes more heavily.

Table 4.16: Unweighted versus balanced class weights (test set)

Class	Recall (unweighted)	Recall (balanced)	F1 (unweighted)	F1 (balanced)
Dropout	0.754	0.662	0.774	0.716
Enrolled	0.333	0.566	0.396	0.448
Graduate	0.898	0.749	0.837	0.785
Accuracy	0.750	0.688		
Macro F1			0.669	0.650

Interpretation. Balanced weighting raises the recall of the Enrolled class from 0.333 to 0.566, but at a cost: Dropout recall falls from 0.754 to 0.662, Graduate recall from 0.898 to 0.749, accuracy from 75.0% to 68.8% and macro F1 slightly from 0.669 to 0.650. Judging by accuracy alone would favour the unweighted model and hide its weakness on Enrolled, whereas judging by Enrolled recall alone would favour the weighted model. Neither is free of trade-offs. The unweighted model was kept as the main model because it has the best accuracy and macro F1, and the weighted version is a sensible option if an institution prefers to catch more Enrolled students at the cost of more false alarms.

7. INTERPRETATION AND RECOMMENDATIONS

7.1 Model Coefficients

In a multinomial logistic regression each class has its own coefficient vector. For the Dropout class, a positive coefficient means that a larger value of the feature raises the log-odds of dropout relative to the other classes, other variables held fixed; an odds ratio (exp of the coefficient) below 1 means lower odds. Numerical variables were standardised, so their coefficients refer to a change of one standard deviation.

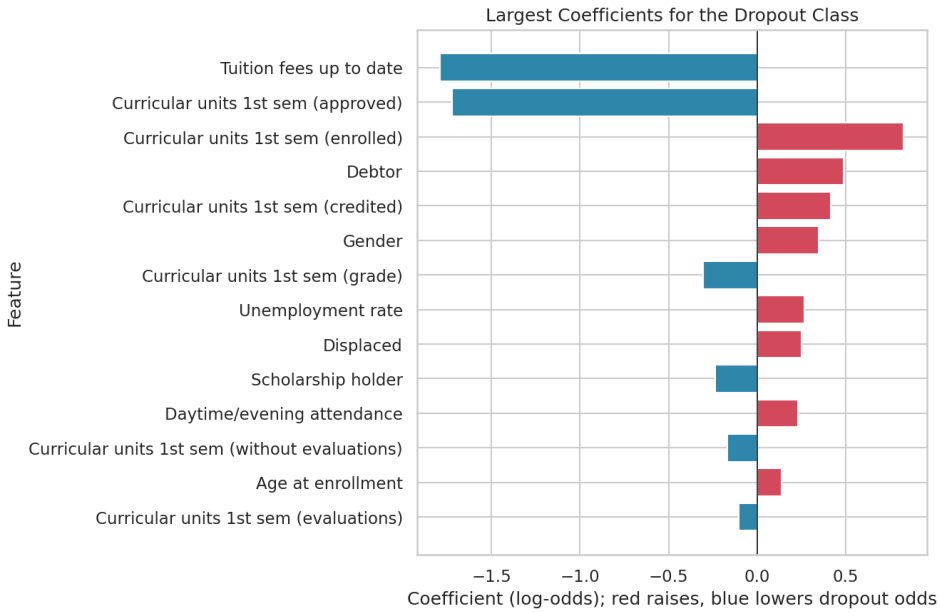


Figure 4.9: Largest logistic regression coefficients for the Dropout class

Table 4.17: Largest Dropout-class coefficients among interpretable features

Feature	Coefficient	Odds ratio
Tuition fees up to date	−1.789	0.167
Sem-1 units approved	−1.721	0.179
Sem-1 units enrolled	+0.828	2.288
Debtor	+0.490	1.632
Sem-1 units credited	+0.414	1.513
Gender (1 = male)	+0.347	1.415
Sem-1 grade	−0.306	0.737
Unemployment rate	+0.267	1.306
Displaced	+0.252	1.287
Scholarship holder	−0.237	0.789

7.2 Variables Associated with the Outcomes

- **Tuition fees up to date** (OR = 0.17) and **units approved in semester 1** (OR = 0.18 per standard deviation) are the strongest variables associated with *lower* odds of dropout. This agrees with the descriptive findings (86.6% against 24.7% dropout by fee status).

- **Debtor status** (OR = 1.63), **male gender** (OR = 1.42), **higher regional unemployment** (OR = 1.31) and **displacement** (OR = 1.29) are associated with higher odds of dropout; **scholarship holding** (OR = 0.79) with lower odds.
- **Units enrolled** and **units credited** in semester 1 have positive coefficients. This is a conditional effect: once approved units are accounted for, enrolling in many units without passing them signals difficulty. These unit counts are strongly correlated with each other, so individual coefficients should not be read in isolation.
- **Age at enrolment** and **admission grade** have smaller coefficients than first-semester and financial variables, consistent with the weak admission-grade differences seen in Chapter 3.

Important caution: these are associations after adjustment for the other variables. They do not show that fees, gender, unemployment or displacement *cause* dropout, and a demographic characteristic must never be presented as the reason a student leaves.

7.3 Limitations of the Model

- **Weak on the Enrolled class** (F1 = 0.396): predictions for students who will still be enrolled are unreliable.
- **Linear and additive:** the model cannot capture interactions (for example, a mature student who is also in debt) or non-linear effects such as the zero-grade spike. Tree-based models might perform better but are harder to explain.
- **Single institution and period:** the data come from one Portuguese institution, so the model may not transfer to other universities, countries or later cohorts.
- **Correlated predictors:** the first-semester unit counts overlap heavily, which makes individual coefficients unstable.
- **Ambiguous zero values:** a first-semester grade of 0 may mean no examinations taken, withdrawal or missing records. This could not be verified.
- **Fairness risk:** gender, age, nationality and parental background act as proxies and could lead to unfair profiling if used carelessly.
- **Moderate accuracy:** about one in four test students is misclassified, so predictions are screening signals and not facts.

7.4 Evidence-Based Recommendations

- **Introduce an early-alert review at the end of the first semester.** Dropouts approve 2.55 units on average against 6.23 for Graduates, and 40.1% have no graded units. Students with very few approved units or no graded units should be contacted for academic advising or tutoring before the second semester begins.
- **Offer proactive financial support.** Dropout is 86.6% where fees are unpaid and 62.0% among debtors, but only 12.2% among scholarship holders. Instalment plans, emergency bursaries and early outreach when a payment is missed are worth investigating, ideally through a pilot with a comparison group to test whether they actually reduce dropout.
- **Provide flexible provision for mature students.** Dropout is 54% to 58% among students aged 25 and above, against 21.0% for those aged 19 or under. Evening or part-time options, recorded lectures and work-compatible timetables should be explored, and these students should be asked directly about the barriers they face.

- **Monitor the "enrolled but struggling" group separately.** Because the model is weak for this class, simple rules (units enrolled but not approved) combined with advisor judgement should be used instead of model output.

7.5 Responsible Use of the Model

- **Support, not punishment:** a high predicted risk should lead to an offer of help, never to restricted access, reduced opportunities or a negative mark on a student's record.
- **Human in the loop:** advisers should review every flag, and students should be informed and able to question or opt out of being flagged.
- **Audit for fairness:** protected characteristics should not be used as grounds for decisions, and error rates should be reported separately for different groups to detect unequal treatment.
- **Treat outputs as probabilities:** because roughly 25% of predictions are wrong, no student should be labelled a "dropout".
- **Avoid causal claims, and re-validate regularly** on new cohorts, tracking whether interventions genuinely improve outcomes.

8. CONCLUSION

This case study analysed the records of 4,424 students to understand dropout and academic success. The data were found to be complete and free of duplicates, but several attributes were integer codes that had to be treated as categories, and a cluster of zero first-semester grades carried real meaning. Exploratory analysis showed that first-semester academic performance, financial status (fees, debt and scholarship) and age at enrolment are most strongly related to the final outcome, whereas admission grade differs only slightly.

A Kruskal-Wallis test, chosen because ANOVA's assumptions did not hold, confirmed that first-semester grades differ significantly across Dropout, Enrolled and Graduate students ($H = 1094.19$, $p < 0.001$) with a large effect size ($\varepsilon^2 = 0.247$). The result was interpreted with care, since part of the gap arises from students who recorded no graded units.

A multinomial logistic regression using only information available by the end of the first semester achieved 75.0% accuracy and a macro F1-score of 0.669. It performs well for Graduate and Dropout students but poorly for Enrolled students, and adding second-semester data improved the macro F1-score only to 0.695. Class weighting improved recall for the Enrolled class but at the cost of accuracy and of the other two classes. The model is best used as a supportive early-warning tool, together with the limitations and safeguards set out in Chapter 7, and not as a basis for causal claims or for penalising students.

OVERALL CONCLUSION

Comparative Summary of the Two Case Studies

The two case studies apply the same data-science workflow (data understanding, preparation, analysis, modelling, evaluation and interpretation) to very different problems. Case Study 1 is unsupervised and looks for structure in unlabelled retail transactions, whereas Case Study 4 is supervised and predicts a known outcome for students. Table C.1 summarises how they compare.

Table C.1: Comparison of Case Study 1 and Case Study 4

Aspect	Case Study 1: Shopping baskets	Case Study 4: Student outcomes
Learning type	Unsupervised (clustering)	Statistical testing and supervised multiclass classification
Data	7,500 transactions, 119 products, binary basket matrix	4,424 students, 36 predictors, target with three classes
Key preparation	Positional columns cleaned into product sets; PCA to 20 components	Attribute types decided by meaning; one-hot encoding, scaling, stratified split; semester-2 data excluded to avoid leakage
Main methods	K-means, Ward, DBSCAN, spectral clustering	Kruskal-Wallis test, multinomial logistic regression
Evaluation	Silhouette, Davies–Bouldin index, coverage	Confusion matrix, precision, recall, F1 (per class and macro-averaged)
Headline result	K-means (K = 5): silhouette 0.1605, DBI 2.0539, 100% coverage; DBSCAN flags 17.56% as noise	Accuracy 75.0%, macro F1 0.669; H = 1094.19, $\varepsilon^2 = 0.247$
Main caveat	Groups are hypotheses; no external ground truth	Associations are not causes; weak on Enrolled class; fairness risk

Common Lessons

- **A single headline number can mislead.** In Case Study 1 the best silhouette belonged to DBSCAN, which left out 17.56% of transactions, so coverage had to be reported with the score. In Case Study 4 an accuracy of 75% hid an F1-score of only 0.396 for the Enrolled class, so macro-averaged and per-class metrics were used.
- **Preparation depends on what the data mean.** The product columns in Case Study 1 were positions and not fields, and the integer codes in Case Study 4 were categories and not quantities. Treating either as ordinary numeric columns would have produced misleading results.
- **Results are evidence for decisions, not decisions in themselves.** The basket groups need validation on later transactions before being called shopping missions, and the dropout model should support, not replace, human advisers.
- **Claims must stay within what the data can show.** Neither dataset supports customer segmentation or causal statements, and both reports say so explicitly.

Concluding Remarks

Together, the two studies show the complete range of the course workflow, from cleaning unconventional data and exploring it visually, to testing hypotheses, building and comparing models, and judging their limits honestly. In both cases the most defensible conclusion was a

measured one: the methods reveal useful and interpretable patterns, but the next step is validation on new data and, where decisions affect people, careful and responsible use.

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